

1. (4 points per part) Parts (a), (b), and (c) are unrelated.

- (a) The two lines $x = 1 + t$, $y = t$, $z = 1 - 2t$ and $x = 1$, $y = 4t$, $z = 1 - 3t$ intersect at the point $(1, 0, 1)$. Find the equation for the plane containing these two intersecting lines.

$$\langle 1, 1, -2 \rangle \times \langle 0, 4, -3 \rangle = \langle 5, 3, 4 \rangle$$

Plane w/ normal vector $\langle 5, 3, 4 \rangle$ through pt $(1, 0, 1)$:

$$5(x-1) + 3y + 4(z-1) = 0$$

equation for plane:

$$5x + 3y + 4z = 9$$

- (b) Find parametric equations for the line of intersection of the two planes $x + y + z = 1$ and $2x + y - 3z = -3$.

$$\begin{aligned} x + y + z &= 1 \\ -(2x + y - 3z &= -3) \\ -x + 4z &= 4 \end{aligned}$$

One point: $x=0, z=1, y=0$

Another: $z=0, x=-4, y=5$

Line through $(0, 0, 1)$ & $(-4, 5, 0)$:

direction $\langle -4, 5, -1 \rangle$

$$x = -4t \quad y = 5t \quad z = 1 - t$$

equations for line:

- (c) Find all points of intersection of the line through $(0, 0, 1)$ and $(3, 4, 1)$ and the paraboloid $100z = x^2 + y^2$.

$$\begin{aligned} \text{Line: } x &= 3t \\ y &= 4t \\ z &= 1 \end{aligned} \quad \left. \begin{array}{l} \text{Plug in: } 100 = (3t)^2 + (4t)^2 \\ 100 = 25t^2 \\ t = \pm 2 \end{array} \right\}$$

Plug back in

intersection point(s): $(x, y, z) =$

$$(-6, -8, 1) \text{ \& } (6, 8, 1)$$

2. (6 points per part) Parts (a) and (b) are unrelated.

- (a) Find a vector function for the curve of intersection between the surface $5x^2 + y^2 - z^2 = 4$ and the plane $z = x$.

$$\begin{aligned} &\downarrow \\ &5x^2 + y^2 - x^2 = 4 \\ &4x^2 + y^2 = 4 \rightarrow \text{ellipse! Can parametrize w/ } \cos t \text{ \& } \sin t. \\ &x = \cos t, \quad y = 2\sin t \text{ works} \\ &z = x = \cos t \end{aligned}$$

$$\vec{r}(t) = \langle \cos t, 2\sin t, \cos t \rangle$$

vector function: _____

- (b) Find the curvature of $\mathbf{r}(t) = \langle t^3, t^2 - 1, 3t + 7 \rangle$ at the point $(-8, 3, 1)$.

$$\begin{aligned} \vec{r}'(t) &= \langle 3t^2, 2t, 3 \rangle & \vec{r}'(-2) &= \langle 12, -4, 3 \rangle & t &= -2 \\ \vec{r}''(t) &= \langle 6t, 2, 0 \rangle & \vec{r}''(-2) &= \langle -12, 2, 0 \rangle \\ \vec{r}'(-2) \times \vec{r}''(-2) &= \langle -6, -36, -24 \rangle \end{aligned}$$

$$|\vec{r}'(-2)| = \sqrt{12^2 + 4^2 + 3^2} = 13$$

$$|\vec{r}''(-2)| = 6\sqrt{1^2 + 6^2 + 4^2} = 6\sqrt{53}$$

$$K = \frac{|\vec{r}'(-2) \times \vec{r}''(-2)|}{|\vec{r}'(-2)|^3}$$

$$K = \frac{6\sqrt{53}}{13^3}$$

curvature: _____

3. (6 points per part) For parts (a) and (b), let $f(x, y) = x^2y^2 - 4x^2 - y^2$.

(a) Find all the saddle points of $f(x, y)$.

$$f_x(x, y) = 2xy^2 - 8x = 0 \rightarrow 2x(y^2 - 4) = 0 \rightarrow x = 0 \text{ or } y = \pm 2$$

$$f_y(x, y) = 2x^2y - 2y = 0 \rightarrow 2y(x^2 - 1) = 0 \rightarrow y = 0 \text{ or } x = \pm 1$$

Five crit. pts: $(0, 0)$, $(1, 2)$, $(-1, 2)$, $(1, -2)$, $(-1, -2)$

$$f_{xx}(x, y) = 2y^2 - 8 \quad D(0, 0) > 0 \leftarrow \text{not a saddle point}$$

$$f_{yy}(x, y) = 2x^2 - 2 \quad D(\pm 1, \pm 2) < 0$$

$$f_{xy}(x, y) = 4xy$$

saddle points:

$$(1, 2), (1, -2), (-1, 2), (-1, -2)$$

(b) Find the equation of the tangent plane to the surface $z = f(x, y)$ at $x = 2, y = 1$ and use it to approximate $f(2.1, 0.8)$.

$$f(2, 1) = -13$$

$$f_x(2, 1) = -12$$

$$f_y(2, 1) = 6$$

$$\text{Tangent plane: } z = -13 - 12(x - 2) + 6(y - 1)$$

$$f(2.1, 0.8) \approx -13 - 12(2.1 - 2) + 6(0.8 - 1) = -15.4$$

tangent plane: $z = -13 - 12(x - 2) + 6(y - 1)$ $f(2.1, 0.8) \approx -15.4$

4. (12 points) Find the absolute maximum and absolute minimum values of the function

$$f(x, y) = x^2 - y^2 + 2y$$

on the region enclosed by $x = 0$, $y = 4$ and $y = x^2$.

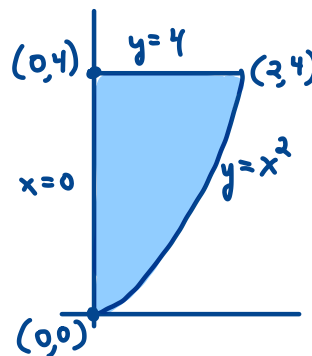
Crit pts:
$$\begin{aligned} f_x(x, y) &= 2x = 0 \\ f_y(x, y) &= -2y + 2 = 0 \end{aligned} \quad \left. \begin{array}{l} x=0 \\ y=1 \end{array} \right\}$$

Boundary

Top: $y=4$: $f(x, y) = x^2 - 8$
 $0 \leq x \leq 2$ $f'(x) = 2x = 0$
 Check $(0, 4)$ & $(2, 4)$ $x=0$

Left: $x=0$: $f(0, y) = -y^2 + 2y$
 $0 \leq y \leq 4$ $f'(y) = -2y + 2 = 0$
 Check $y=1$
 $(0, 0), (0, 1), (0, 4)$

Bottom-right: $f(x, x^2) = x^2 - x^4 + 2x^2$
 $f'(x) = 6x - 4x^3 = 0$
 $= 2x(3 - 2x^2) = 0$
 Check $(0, 0)$, $(2, 4)$, $(\sqrt{\frac{3}{2}}, \frac{3}{2})$
 $x=0, x=\sqrt{\frac{3}{2}}, x=\sqrt{\frac{3}{2}}$
 not in domain



Points to check:

$$f(0, 1) = 1$$

$$f(0, 4) = -8$$

$$f(2, 4) = -4$$

$$f(0, 0) = 0$$

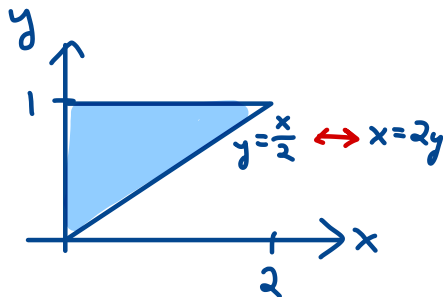
$$f\left(\sqrt{\frac{3}{2}}, \frac{3}{2}\right) = \frac{9}{4}$$

absolute extrema:

$$\min -8, \max \frac{9}{4}$$

5. (12 points) Reverse the order of integration and evaluate

$$\int_0^2 \int_{x/2}^1 \frac{x}{y^3 + 1} dy dx.$$



$$= \int_0^1 \int_0^{2y} \frac{x}{y^3 + 1} dx dy$$

$$= \int_0^1 \left(\frac{\frac{1}{2} x^2}{y^3 + 1} \right) \Bigg|_{x=0}^{x=2y} dy$$

$$= \int_0^1 \frac{2y^2}{y^3 + 1} dy \quad \begin{array}{l} u = y^3 + 1 \\ du = 3y^2 dy \end{array}$$

$$= \int_1^2 \frac{\frac{2}{3}}{u} du = \frac{2}{3} \ln|u| \Bigg|_1^2$$

$$= \frac{2}{3} (\ln(2) - 0)$$

answer =

$$\boxed{\frac{2}{3} \ln(2)}$$

6. (12 points) Let R be the region inside the circle $x^2 + y^2 = 4y$, outside the circle $x^2 + y^2 = 8$, and in the first quadrant (shown below). Evaluate

$$\iint_R \frac{x}{x^2 + y^2} dA$$

Convert to polar:

$$x^2 + y^2 = 4y \rightarrow r^2 = 4r \sin \theta \rightarrow r = 4 \sin \theta$$

$$x^2 + y^2 = 8 \rightarrow r = \sqrt{8}$$

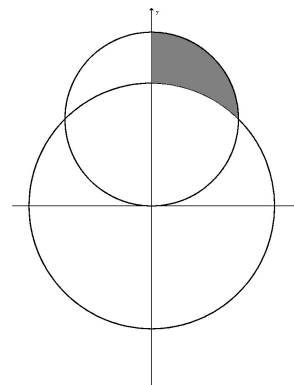
Intersection: $\sqrt{8} = 4 \sin \theta \rightarrow \sin \theta = \frac{\sqrt{2}}{2} \rightarrow \theta = \frac{\pi}{4}$

$$\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \int_{\sqrt{8}}^{4 \sin \theta} \frac{\cancel{r} \cos \theta}{\cancel{r}} dr d\theta = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos \theta \left[r \right]_{r=\sqrt{8}}^{r=4 \sin \theta} d\theta$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (4 \sin \theta \cos \theta - \sqrt{8} \cos \theta) d\theta$$

$$= \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (2 \sin(2\theta) - \sqrt{8} \cos \theta) d\theta = \left(-\cos 2\theta - \sqrt{8} \sin \theta \right) \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}} = (1 - \sqrt{8}) - (0 - 2)$$

$$= 3 - \sqrt{8}$$



$$= \boxed{3 - \sqrt{8}}$$

answer = _____

7. For parts (a)–(c), let $f(x) = \ln(2x - 1)$.

(a) (5 points) Find the second Taylor polynomial, $T_2(x)$, for $f(x)$ based at $b = 1$.

$$f(x) = \ln(2x-1) \quad f(1) = 0$$

$$f'(x) = \frac{2}{2x-1} \quad f'(1) = 2$$

$$f''(x) = \frac{-4}{(2x-1)^2} \quad f''(1) = -4$$

$$T_2(x) = 2(x-1) - 2(x-1)^2$$

(b) (4 points) Use your answer to part (a) to approximate $\ln(1.1)$.

$$\begin{aligned} \ln(1.1) &= \ln(2.1-1) = f(1.05) \approx T_2(1.05) \\ &= 2(1.05-1) - 2(1.05-1)^2 = 0.1 - 0.005 \end{aligned}$$

$$\ln(1.1) \approx 0.095$$

(c) (5 points) Find an upper bound (as sharp as possible) on the error for your answer from part (b).

$$f'''(x) = \frac{16}{(2x-1)^3}$$

max when x is smallest, at $x=1$ on interval $[1, 1.05]$.

$$\rightarrow \text{Use } M = \frac{16}{1^3} = 16$$

$$|f(x) - T_2(x)| \leq \frac{1}{6}(16)(0.05)^3 = \frac{1}{3000}$$

(Ok to instead use interval $[0.95, 1.05]$,
giving $M = \frac{16}{.9^3}$, error ≤ 0.000457)

$$\text{Error bound: } \frac{1}{3000} = 0.000\bar{3}$$

8. For this problem, you may use the following basic Taylor series:

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k, \quad e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}, \quad \sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}, \quad \cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}$$

- (a) (6 points) Find the Taylor series for $f(x) = \int_0^x e^{2t^2} dt$ based at $b = 0$. Express your answer using $\sum$ -notation.

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!} \quad \text{replace } x \text{ by } 2t^2$$

$$e^{2t^2} = \sum_{k=0}^{\infty} \frac{2^k t^{2k}}{k!}$$

$$\int_0^x e^{2t^2} dt = \sum_{k=0}^{\infty} \left(\int_0^x \frac{2^k t^{2k}}{k!} dt \right)$$

$$\sum_{k=0}^{\infty} \frac{2^k x^{2k+1}}{(2k+1)k!}$$

Taylor series: _____

- (b) (3 points) Find the open interval of convergence for the series you found in (a).

e^x converges for all x
None of these transformations change that.

Interval of convergence: _____

$$(-\infty, \infty)$$

- (c) (5 points) Find $f^{(2023)}(0)$, i.e. the 2023rd derivative of f at 0.

x^{2023} term of $\sum_{k=0}^{\infty} \frac{2^k x^{2k+1}}{(2k+1)k!}$ is at $2k+1=2023 \rightarrow k=1011 \rightarrow \frac{2^{1011} x^{2023}}{2023 \cdot 1011!}$

$$\frac{f^{(2023)}(0) x^{2023}}{2023!} = \frac{2^{1011} x^{2023}}{2023 \cdot 1011!} \rightarrow f^{(2023)}(0) = \frac{2023! 2^{1011}}{2023 \cdot 1011!}$$

$$\frac{2022! \cdot 2^{1011}}{1011!}$$

$f^{(2023)}(0) =$ _____